

YEAR 7 MATHEMATICS

Chapter 7

Decimals

WORKSHEETS

7

How to use this booklet

1 There is **one worksheet for each lesson**. It has three columns that get harder from left to right: **Easy**, then **Medium**, then **Challenging**.

WE DO **Top of each column.** Your teacher works the examples on the board. **Copy every step** into the space.

2 Read the **speech bubble** at the top of each column: it tells you what to do. A new bubble means a new kind of question.

YOU DO **The rest of the column.** Your turn. Column 1: answer on the line. Columns 2 and 3: show your working on the lines.

Contents and progress tracker

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In this chapter you will

- order decimals
- convert between decimals and fractions
- add, subtract, multiply and divide decimals
- multiply and divide by powers of 10
- round decimals and write recurring decimals
- solve money problems

About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \rightarrow z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is a double spiral deep inside Seahorse Valley. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

Easy

Questions 1–29

Which decimal is larger?
Compare the digits place by place from the left.



WE DO With your teacher: copy the working.

E1 7.48 or 7.9

E2 9.59 or 9.56

YOU DO Your turn.

1 9.14 or 9.12

9.14

2 2.23 or 3.13

3 9.16 or 9.14

4 3.69 or 3.78

5 1.49 or 1.8

6 8.85 or 8.82

7 4.26 or 4.32

8 1.22 or 1.1

9 9.92 or 9.8

10 5.18 or 5.7

11 3.31 or 3.27

12 2.8 or 3.3

13 5.55 or 5.56

14 5.9 or 5.89

15 1.22 or 1.27

16 8.95 or 9

17 8.91 or 9.81

18 4.66 or 5.56

19 7.23 or 7.2

20 6.66 or 6.71

21 7.21 or 7.5

22 8.84 or 8.88

23 9.43 or 9.4

24 2.21 or 2.25

25 4.93 or 4.1

26 5.46 or 5.2

27 5.73 or 5.76

28 9.96 or 10.56

29 8.4 or 8.37

Medium

Questions 30–39

Write the decimals in ascending order. Add zeros so they all have the same number of places.



WE DO With your teacher: copy the working.

E1 4.5, 4.38, 4.35, 4.24

E2 4.5, 4.47, 4.673, 4.27

YOU DO Your turn.

30 5.7, 5.88, 5.825, 5.36

5.700, 5.880, 5.825, 5.360

5.36, 5.7, 5.825, 5.88

31 3.2, 3.22, 3.635, 3.9

3.200, 3.220, 3.635, 3.900

32 1.1, 1.52, 1.271, 1.56

33 3.3, 3.36, 3.927, 3.78

34 3.5, 3.56, 3.277, 3.49

35 4.3, 4.14, 4.39, 4.91

36 2.5, 2.24, 2.293, 2.58

37 5.4, 5.06, 5.236, 5.75

38 5.6, 5.54, 5.553, 5.9

39 0.6, 0.73, 0.948, 0.74

Challenging

Questions 40–47

Write a decimal that fits, then explain how you know.



WE DO With your teacher: copy the working.

E1 Write a decimal between 2.5 and 2.6.

E2 Write a decimal between 1.8 and 1.9.

YOU DO Your turn.

40 Write a decimal between 5.6 and 5.7.

5.60 and 5.70

e.g. ___ is between them

41 Write a decimal between 3.1 and 3.2.

___ and ___

e.g. ___ is between them

42 Write a decimal between 4.6 and 4.7.

43 Write a decimal between 7.6 and 7.7.

44 Write a decimal between 7.2 and 7.3.

45 Write a decimal between 8.2 and 8.3.

46 Write a decimal between 6.3 and 6.4.

47 Write a decimal between 5.5 and 5.6.

Easy

Questions 1–29

Write each decimal as a fraction with 10, 100 or 1000 on the bottom.



WE DO With your teacher: copy the working.

E1 0.317

E2 0.754

YOU DO Your turn.

1 0.628

$\frac{628}{1000}$

2 0.52

3 0.2

4 0.49

5 0.39

6 0.8

7 0.456

8 0.7

9 0.98

10 0.874

11 0.1

12 0.4

13 0.781

14 0.873

15 0.558

16 0.213

17 0.08

18 0.28

19 0.3

20 0.977

21 0.67

22 0.6

23 0.46

24 0.48

25 0.858

26 0.154

27 0.45

28 0.37

29 0.623

Medium

Questions 30–45



Write as a fraction, then simplify fully.

WE DO With your teacher: copy the working.

E1 0.08

E2 0.72

YOU DO Your turn.

30 0.75

$= \frac{75}{100}$

$= \frac{3}{4}$

31 0.46

$= \frac{46}{100}$

$= \frac{23}{50}$

32 0.82

$= \frac{82}{100}$

$= \frac{41}{50}$

33 0.88

34 0.05

35 0.85

36 0.78

37 0.06

38 0.45

39 0.55

40 0.22

41 0.16

42 0.66

43 0.14

44 0.56

45 0.92

Challenging

Questions 46–52

Write as a mixed numeral in simplest form.



WE DO With your teacher: copy the working.

E1 8.16

E2 4.02

YOU DO Your turn.

46 4.54

$= 4 \text{ and } \frac{54}{100}$

$= \frac{27}{50}$

47 2.94

$= \frac{94}{100}$

$= \frac{47}{50}$

48 5.64

49 2.45

50 6.36

51 2.88

52 9.16

Easy

Questions 1–30

Add or subtract. Line up the decimal points.



WE DO With your teacher: copy the working.

E1 $7.3 - 5.8$ _____

E2 $7.7 - 6.5$ _____

YOU DO Your turn.

1 $3.2 + 4.6$ _____

7.8

2 $5.1 - 1.4$ _____

3 $5.7 - 5.4$ _____

4 $7.5 + 6.2$ _____

5 $6.9 - 5$ _____

6 $6.2 + 7.5$ _____

7 $6.6 + 2$ _____

8 $8.8 + 4.8$ _____

9 $8.1 + 2.7$ _____

10 $8.1 + 1.5$ _____

11 $6.1 + 8.5$ _____

12 $7.7 + 5.8$ _____

13 $9.9 - 5.6$ _____

14 $7.3 - 4.2$ _____

15 $7.9 + 8.2$ _____

16 $8.8 + 3.6$ _____

17 $6.1 + 7.3$ _____

18 $7.1 - 6.4$ _____

19 $6.9 + 7.1$ _____

20 $8.7 + 3.2$ _____

21 $9.2 - 8.8$ _____

22 $5.8 + 5.6$ _____

23 $4.7 + 6.9$ _____

24 $5.8 + 4.7$ _____

25 $8.9 + 2.2$ _____

26 $6.3 + 8.5$ _____

27 $8.3 + 9.2$ _____

28 $4.9 - 1.1$ _____

29 $8.8 + 2.2$ _____

30 $6.5 + 7.3$ _____

Medium

Questions 31–52

Fill empty places with zeros, line up the points, then add or subtract.



WE DO With your teacher: copy the working.

E1 $4.9 - 3.79$ _____

E2 $69 - 3.94$ _____

YOU DO Your turn.

31 $7.3 - 1.08$ _____

32 $9.7 - 4.15$ _____

$7.30 - 1.08$

$9.70 - 4.15$

$= 6.22$

$= \underline{\quad}$

33 $9.7 + 9.91$ _____

34 $30 + 2.03$ _____

$\underline{\quad} + \underline{\quad}$

$= \underline{\quad}$

35 $28 + 6.44$ _____

36 $25 + 4.11$ _____

37 $80 - 1.22$ _____

38 $2.6 + 6.97$ _____

39 $9.5 - 1.31$ _____

40 $53 + 4.5$ _____

41 $32 + 6.65$ _____

42 $10 - 7.2$ _____

43 $46 - 8.38$ _____

44 $9.2 - 8.62$ _____

45 $3.3 + 1.67$ _____

46 $4.3 + 7.29$ _____

47 $22 - 1.71$ _____

48 $9 - 5.12$ _____

49 $20 + 4.44$ _____

50 $6.6 + 4.15$ _____

51 $4.7 + 2.34$ _____

52 $81 - 9.17$ _____

Challenging

Questions 53–59

Write a number sentence, then answer the question.



WE DO With your teacher: copy the working.

E1 Mia runs 3.26 km and then 4.39 km. How far does she run altogether?

E2 A 12 m rope has 3.75 m cut off. How much is left?

YOU DO Your turn.

53 Mia runs 8.78 km and then 6.18 km. How far does she run altogether?

$8.78 + 6.18$

$= \underline{\quad} \text{ km}$

54 A 15 m rope has 4.77 m cut off. How much is left?

$\underline{\quad} - \underline{\quad}$

$= \underline{\quad} \text{ m}$

55 Mia runs 3.78 km and then 2.38 km. How far does she run altogether?

56 A 11 m rope has 2.82 m cut off. How much is left?

57 Mia runs 3.68 km and then 1.63 km. How far does she run altogether?

58 A 10 m rope has 3.35 m cut off. How much is left?

59 Mia runs 4.07 km and then 8.07 km. How far does she run altogether?

Easy

Questions 1–24

Multiply by 10, 100 or 1000: the digits move left one place for each zero.

WE DO With your teacher: copy the working.

E1 5.15×100 _____

E2 0.769×10 _____

YOU DO Your turn.

1 56.6×100 5 660

2 0.666×10 _____

3 0.34×100 _____

4 85.5×100 _____

5 2.26×10 _____

6 0.674×10 _____

7 15×10 _____

8 8.69×10 _____

9 $6.71 \times 1\,000$ _____

10 7.32×10 _____

11 2.39×10 _____

12 42.7×10 _____

Divide by 10, 100 or 1000: the digits move right one place for each zero.

WE DO With your teacher: copy the working.

E3 $583.3 \div 1\,000$ _____

YOU DO Your turn.

13 $882.1 \div 1\,000$ 0.8821

14 $172.7 \div 1\,000$ _____

15 $352.2 \div 10$ _____

16 $3\,943 \div 100$ _____

17 $9\,593 \div 10$ _____

18 $416.8 \div 10$ _____

19 $1\,005 \div 10$ _____

20 $5\,597 \div 100$ _____

21 $9\,489 \div 100$ _____

22 $4\,958 \div 100$ _____

23 $3\,375 \div 1\,000$ _____

24 $5\,844 \div 1\,000$ _____

Medium

Questions 25–33

Find the missing power of 10. Count how many places the digits moved.

WE DO With your teacher: copy the working.

E1 $7.84 \div \square = 0.00784$

E2 $4.68 \times \square = 4\,680$

YOU DO Your turn.

25 $4.98 \div \square = 0.498$

moved 1 places right

$\square = 10$

26 $3.5 \div \square = 0.0035$

moved 3 places right

$\square =$ _____

27 $2.72 \times \square = 2\,720$

moved ___ places left

$\square =$ _____

28 $4.53 \div \square = 0.453$

29 $0.47 \div \square = 0.00047$

30 $6.23 \times \square = 6\,230$

31 $9.39 \div \square = 0.0939$

32 $0.29 \times \square = 29$

33 $0.66 \div \square = 0.0066$

Challenging

Questions 34–41

Change the units by multiplying or dividing by a power of 10.

WE DO With your teacher: copy the working.

E1 99.9 m to centimetres

E2 203 g to kilograms

E3 71.3 cm to millimetres

YOU DO Your turn.

34 88.3 m to centimetres

1 m = 100 cm, so $\times 100$

_____ $\times$ _____ = _____ cm

35 343 g to kilograms

_____ kg = _____ g, so $\div$ _____

_____ $\div$ _____ = _____ kg

36 50.2 cm to millimetres

37 35.3 m to centimetres

38 495 g to kilograms

39 9.5 cm to millimetres

40 29.8 m to centimetres

41 157 g to kilograms

Easy

Questions 1–24

Use $6 \times 7 = 42$. Count the decimal places in the question: the answer has the same number.

WE DO With your teacher: copy the working.

E1 6×0.07 _____

E2 0.6×0.7 _____

YOU DO Your turn.

1 0.06×0.07 0.0042

2 6×0.7 _____

3 0.6×0.07 _____

4 0.06×0.7 _____

Use your times tables and count the decimal places.

WE DO With your teacher: copy the working.

E3 0.6×0.06 _____

YOU DO Your turn.

5 6×0.4 2.4

6 11×0.05 _____

7 0.6×0.09 _____

8 0.2×0.4 _____

9 0.6×0.6 _____

10 3×0.06 _____

11 0.4×0.5 _____

12 0.4×0.05 _____

13 0.8×0.02 _____

14 0.3×0.4 _____

15 0.7×0.09 _____

16 11×0.02 _____

17 0.3×0.9 _____

18 10×0.5 _____

19 2×0.7 _____

20 1.1×0.3 _____

21 11×0.3 _____

22 1×0.04 _____

23 4×0.02 _____

24 1×0.05 _____

Medium

Questions 25–33

Use the fact given. Count the decimal places.

WE DO With your teacher: copy the working.

E1 $21 \times 48 = 1\,008$. Find 2.1×4.8 .

E2 $33 \times 34 = 1\,122$. Find 0.33×3.4 .

YOU DO Your turn.

25 $28 \times 19 = 532$. Find 0.28×1.9 .

3 decimal places

= 0.532

26 $16 \times 16 = 256$. Find 1.6×16 .

1 decimal places

= ____

27 $29 \times 28 = 812$. Find 0.29×2.8 .

____ decimal places

= ____

28 $31 \times 15 = 465$. Find 3.1×1.5 .

29 $17 \times 47 = 799$. Find 1.7×47 .

30 $39 \times 18 = 702$. Find 3.9×18 .

31 $33 \times 47 = 1\,551$. Find 3.3×47 .

32 $24 \times 36 = 864$. Find 2.4×36 .

33 $15 \times 28 = 420$. Find 0.15×28 .

Challenging

Questions 34–39

Use the fact given to find each answer. Explain your thinking.

WE DO With your teacher: copy the working.

E1 $25 \times 45 = 1\,125$. Find $112.5 \div 45$.

E2 $13 \times 17 = 221$. Find $2.21 \div 17$.

YOU DO Your turn.

34 $36 \times 22 = 792$. Find $7.92 \div 22$.

$792 \div 22 = 36$

7.92 is $792 \div 100$, so the answer is $36 \div 100$

= ____

35 $13 \times 26 = 338$. Find $33.8 \div 26$.

____ $\div$ ____ = ____

____ is ____ $\div$ ____, so the answer is ____ $\div$ ____

= ____

36 $34 \times 17 = 578$. Find $57.8 \div 17$.

37 $48 \times 48 = 2\,304$. Find $230.4 \div 48$.

38 $42 \times 44 = 1\,848$. Find $184.8 \div 44$.

39 $43 \times 16 = 688$. Find $6.88 \div 16$.

Easy

Questions 1–30

Multiply the digits, then put the decimal point back.



WE DO With your teacher: copy the working.

E1 9.3×5 _____

E2 3.5×2 _____

YOU DO Your turn.

1 3.8×7 _____ **26.6**

2 5.3×5 _____

3 4×6 _____

4 7.1×3 _____

5 3.2×9 _____

6 2.3×7 _____

7 2.7×5 _____

8 8×6 _____

9 3.8×6 _____

10 9.6×7 _____

11 9.2×7 _____

12 9×4 _____

13 9×7 _____

14 7.9×9 _____

15 7.9×4 _____

16 3.4×8 _____

17 3.6×6 _____

18 5.5×9 _____

19 9.4×9 _____

20 6.9×7 _____

21 3.4×6 _____

22 8.3×7 _____

23 1.7×6 _____

24 3.2×5 _____

25 8.2×2 _____

26 7.9×2 _____

27 7.6×8 _____

28 5.9×2 _____

29 8.9×7 _____

30 1.6×8 _____

Medium

Questions 31–40

Ignore the points, multiply, then count the decimal places.

WE DO With your teacher: copy the working.

E1 0.42×0.6 _____

E2 0.3×0.5 _____

YOU DO Your turn.

31 0.12×0.3 _____

$12 \times 3 = 36$

3 decimal places: 0.036

32 0.57×0.9 _____

$57 \times 9 = 513$

___ decimal places: ___

33 0.86×0.9 _____

___ $\times$ ___ = ___

___ decimal places: ___

34 0.9×0.6 _____

35 0.57×0.6 _____

36 0.33×0.6 _____

37 0.14×0.2 _____

38 0.78×0.3 _____

39 0.25×0.8 _____

40 0.52×0.8 _____

Challenging

Questions 41–47

Estimate first, then multiply.



WE DO With your teacher: copy the working.

E1 8.47×9.5 _____

E2 3.74×2.1 _____

YOU DO Your turn.

41 7.74×7.5 _____

estimate: $8 \times 8 = 64$

$774 \times 75 = 58\ 050$

___ decimal places: ___

42 8.48×7.4 _____

estimate: ___ $\times$ ___ = ___

___ $\times$ ___ = ___

___ decimal places: ___

43 3.11×7.6 _____

44 6.74×7.9 _____

45 2.87×7.9 _____

46 1.19×6.9 _____

47 6.24×1.6 _____

Easy

Questions 1–30

Divide. Keep the decimal point in the same place.



WE DO With your teacher: copy the working.

E1 $13.3 \div 7$

E2 $9.6 \div 6$

YOU DO Your turn.

1 $15.5 \div 5$

3.1

2 $21.6 \div 4$

3 $12 \div 4$

4 $37.6 \div 4$

5 $23.5 \div 5$

6 $40 \div 5$

7 $29.7 \div 9$

8 $27.9 \div 3$

9 $9.6 \div 2$

10 $68.6 \div 7$

11 $79.2 \div 8$

12 $17.1 \div 9$

13 $32 \div 8$

14 $33.6 \div 7$

15 $20 \div 8$

16 $8.4 \div 7$

17 $32.8 \div 8$

18 $51.6 \div 6$

19 $15.6 \div 6$

20 $10.5 \div 7$

21 $36.5 \div 5$

22 $5.4 \div 3$

23 $28.8 \div 3$

24 $7.8 \div 6$

25 $10.5 \div 3$

26 $7.5 \div 5$

27 $32.4 \div 9$

28 $53.1 \div 9$

29 $33 \div 6$

30 $33.2 \div 4$

Medium

Questions 31–52

Use short division. Add zeros after the last digit if you need to.



WE DO With your teacher: copy the working.

E1 $4.6 \div 2$

E2 $7 \div 5$

YOU DO Your turn.

31 $3.8 \div 2$

32 $7.2 \div 8$

$3.800 \div 2$

$7.200 \div 8$

$= 1.9$

$= \underline{\hspace{1cm}}$

33 $1.7 \div 2$

34 $8.4 \div 8$

$\underline{\hspace{1cm}} \div \underline{\hspace{1cm}}$

$= \underline{\hspace{1cm}}$

35 $5.4 \div 4$

36 $4.7 \div 2$

37 $3.8 \div 4$

38 $4.9 \div 8$

39 $3.9 \div 8$

40 $2.6 \div 4$

41 $4.1 \div 2$

42 $2.3 \div 8$

43 $2.2 \div 8$

44 $1.4 \div 8$

45 $1.3 \div 5$

46 $3.5 \div 5$

47 $2.7 \div 2$

48 $5.8 \div 5$

49 $6.7 \div 4$

50 $3.6 \div 4$

51 $6.3 \div 4$

52 $9 \div 5$

Challenging

Questions 53–58

Share the money equally. Write a number sentence.



WE DO With your teacher: copy the working.

E1 \$16.30 is shared equally between 5 people. How much does each get?

E2 \$58.92 is shared equally between 4 people. How much does each get?

YOU DO Your turn.

53 \$120.40 is shared equally between 7 people. How much does each get?

$\$120.40 \div 7$

$= \underline{\hspace{1cm}} \text{ each}$

54 \$44.50 is shared equally between 5 people. How much does each get?

$\underline{\hspace{1cm}} \div \underline{\hspace{1cm}}$

$= \underline{\hspace{1cm}} \text{ each}$

55 \$127.84 is shared equally between 8 people. How much does each get?

56 \$69.72 is shared equally between 4 people. How much does each get?

57 \$9.00 is shared equally between 5 people. How much does each get?

58 \$81.30 is shared equally between 5 people. How much does each get?

Easy

Questions 1–29

Multiply both numbers by 10 so you divide by a whole number.



WE DO

With your teacher: copy the working.

E1 $22.5 \div 0.9$ _____

E2 $23.4 \div 0.9$ _____

YOU DO

Your turn.

1 $19.6 \div 0.7$ _____

28

2 $1.6 \div 0.2$ _____

3 $3.6 \div 0.4$ _____

4 $18.2 \div 0.7$ _____

5 $9.6 \div 0.6$ _____

6 $3.5 \div 0.7$ _____

7 $7.8 \div 0.3$ _____

8 $8.1 \div 0.3$ _____

9 $7.7 \div 0.7$ _____

10 $18.9 \div 0.7$ _____

11 $1.5 \div 0.5$ _____

12 $18.4 \div 0.8$ _____

13 $13.6 \div 0.8$ _____

14 $24.3 \div 0.9$ _____

15 $17.1 \div 0.9$ _____

16 $7 \div 0.7$ _____

17 $8.4 \div 0.4$ _____

18 $18 \div 0.6$ _____

19 $7.8 \div 0.6$ _____

20 $1.6 \div 0.4$ _____

21 $4.8 \div 0.6$ _____

22 $11 \div 0.5$ _____

23 $25.2 \div 0.9$ _____

24 $15.6 \div 0.6$ _____

25 $19.8 \div 0.9$ _____

26 $13 \div 0.5$ _____

27 $7.5 \div 0.3$ _____

28 $16.8 \div 0.7$ _____

29 $1.4 \div 0.7$ _____

Medium

Questions 30–51



Move both decimal points the same number of places, then divide.

WE DO

With your teacher: copy the working.

E1 $4.05 \div 0.5$ _____

E2 $4.3 \div 0.5$ _____

YOU DO

Your turn.

30 $1.32 \div 0.3$

$= 13.2 \div 3$

$= 4.4$

32 $5.94 \div 0.9$

$= ___ \div ___$

$= ___$

34 $5.44 \div 0.8$ _____

36 $4.2 \div 0.7$ _____

38 $2.87 \div 0.7$ _____

40 $6.65 \div 0.7$ _____

42 $0.6 \div 0.5$ _____

44 $4.5 \div 0.9$ _____

46 $2.28 \div 0.6$ _____

48 $4.2 \div 0.5$ _____

50 $0.44 \div 0.2$ _____

31 $2.25 \div 0.9$

$= 22.5 \div 9$

$= ___$

33 $2.61 \div 0.3$ _____

35 $2.7 \div 0.3$ _____

37 $1.35 \div 0.9$ _____

39 $5.81 \div 0.7$ _____

41 $1.3 \div 0.5$ _____

43 $1.83 \div 0.3$ _____

45 $0.54 \div 0.2$ _____

47 $1.26 \div 0.6$ _____

49 $6.64 \div 0.8$ _____

51 $0.84 \div 0.6$ _____

Challenging

Questions 52–56

Divide by a decimal to answer the question.



WE DO

With your teacher: copy the working.

E1 A bottle holds 0.25 L. How many bottles can be filled from 4.5 L?

E2 A bottle holds 0.25 L. How many bottles can be filled from 1 L?

YOU DO

Your turn.

52 A bottle holds 0.25 L. How many bottles can be filled from 2 L?

$2 \div 0.25$

$= 200 \div 25$

$= ___ \text{ bottles}$

53 A bottle holds 0.5 L. How many bottles can be filled from 8 L?

$___ \div ___$

$= ___ \div ___$

$= ___ \text{ bottles}$

54 A bottle holds 0.5 L. How many bottles can be filled from 3 L?

55 A bottle holds 1.25 L. How many bottles can be filled from 18.75 L?

56 A bottle holds 1.5 L. How many bottles can be filled from 36 L?

Easy

Questions 1–29

Write each fraction as a decimal. Make the bottom 10 or 100 first.



WE DO

With your teacher: copy the working.

E1 $\frac{18}{25}$

E2 $\frac{1}{10}$

YOU DO

Your turn.

1 $\frac{21}{50}$

0.42

2 $\frac{2}{5}$

3 $\frac{1}{50}$

4 $\frac{3}{4}$

5 $\frac{7}{10}$

6 $\frac{1}{2}$

7 $\frac{3}{5}$

8 $\frac{1}{4}$

9 $\frac{1}{5}$

10 $\frac{3}{25}$

11 $\frac{11}{20}$

12 $\frac{21}{25}$

13 $\frac{17}{25}$

14 $\frac{9}{10}$

15 $\frac{19}{20}$

16 $\frac{3}{50}$

17 $\frac{13}{20}$

18 $\frac{16}{25}$

19 $\frac{27}{50}$

20 $\frac{1}{25}$

21 $\frac{39}{50}$

22 $\frac{17}{20}$

23 $\frac{4}{5}$

24 $\frac{3}{10}$

25 $\frac{9}{20}$

26 $\frac{9}{25}$

27 $\frac{41}{50}$

28 $\frac{23}{50}$

29 $\frac{1}{20}$

Medium

Questions 30–49



Divide the top by the bottom using short division.

WE DO

With your teacher: copy the working.

E1 $\frac{7}{16}$

E2 $\frac{1}{8}$

YOU DO

Your turn.

30 $\frac{27}{40}$

31 $\frac{1}{16}$

$27.0000 \div 40$

$1.0000 \div 16$

$= 0.675$

$= \underline{\hspace{1cm}}$

32 $\frac{5}{16}$

33 $\frac{11}{16}$

$\underline{\hspace{1cm}} \div \underline{\hspace{1cm}}$

$= \underline{\hspace{1cm}}$

34 $\frac{37}{40}$

35 $\frac{21}{40}$

36 $\frac{5}{8}$

37 $\frac{13}{16}$

38 $\frac{17}{40}$

39 $\frac{3}{8}$

40 $\frac{9}{16}$

41 $\frac{15}{16}$

42 $\frac{33}{40}$

43 $\frac{7}{8}$

44 $\frac{29}{40}$

45 $\frac{39}{40}$

46 $\frac{23}{40}$

47 $\frac{1}{40}$

48 $\frac{11}{40}$

49 $\frac{13}{40}$

Challenging

Questions 50–57

Write as a recurring decimal. Put a dot over the digit (or digits) that repeat.



WE DO

With your teacher: copy the working.

E1 $\frac{1}{3}$

E2 $\frac{2}{3}$

YOU DO

Your turn.

50 $\frac{1}{9}$

$1 \div 9 = 0.111\dots$

$= \underline{\hspace{1cm}}$

51 $\frac{5}{9}$

$\underline{\hspace{1cm}} \div \underline{\hspace{1cm}} = \underline{\hspace{1cm}}\dots$

$= \underline{\hspace{1cm}}$

52 $\frac{1}{6}$

53 $\frac{5}{6}$

54 $\frac{7}{9}$

55 $\frac{1}{11}$

56 $\frac{4}{3}$

57 $\frac{2}{11}$

Easy

Questions 1–29

Round to 1 decimal place.
Look at the second decimal digit.



WE DO With your teacher: copy the working.

E1 77.26

E2 89.54

YOU DO Your turn.

1 81.11

81.1

2 70.14

3 42.91

4 1.19

5 13.78

6 33.56

7 65.34

8 48.04

9 19.47

10 40.02

11 86.06

12 36.04

13 74.49

14 86.43

15 25.48

16 69.16

17 27.71

18 37.39

19 47.65

20 57.98

21 27.21

22 46.24

23 99.94

24 15.09

25 97.07

26 34.57

27 71.59

28 16.46

29 90.03

Medium

Questions 30–39

Round to the number of places given. Circle the digit you round to, then look at the next digit.

WE DO With your teacher: copy the working.

E1 22.0996 (1 d.p.)

E2 91.2065 (2 d.p.)

YOU DO Your turn.

30 12.8775 (3 d.p.)

next digit is 5: round up

= 12.878

31 97.3245 (2 d.p.)

next digit is 4: round down

= 97.32

32 84.5044 (3 d.p.)

next digit is 4: round down

= 84.504

33 12.0799 (1 d.p.)

34 26.1622 (2 d.p.)

35 98.3739 (1 d.p.)

36 30.0885 (2 d.p.)

37 59.17 (3 d.p.)

38 63.2249 (1 d.p.)

39 33.5389 (1 d.p.)

Challenging

Questions 40–49

Use your calculator, then round the answer to 2 decimal places.



WE DO With your teacher: copy the working.

E1 $31 \div 7$

E2 $13 \div 3$

YOU DO Your turn.

40 $42 \div 13$

calculator: 3.23077...

= 3.23 (2 d.p.)

41 $19 \div 7$

calculator: 2.714285...

= 2.71 (2 d.p.)

42 $17 \div 8$

43 $38 \div 7$

44 $50 \div 13$

45 $29 \div 8$

46 $35 \div 13$

47 $24 \div 13$

48 $39 \div 8$

49 $17 \div 13$

Easy

Questions 1–25

Add the prices.

WE DO With your teacher: copy the working.

E1 $\$1.18 + \1.31

E2 $\$12.01 + \17.63

YOU DO Your turn.

1 $\$3.56 + \16.28 $\$19.84$

2 $\$15.45 + \8.29

3 $\$16.67 + \14.39

4 $\$4.88 + \12.73

5 $\$8.82 + \18.71

6 $\$13.38 + \1.40

7 $\$2.03 + \15.21

8 $\$3.30 + \4.75

9 $\$15.34 + \15.37

10 $\$8.32 + \4.27

11 $\$8.13 + \19.93

12 $\$13.58 + \15.30

Find the change. Count up from the price.

WE DO With your teacher: copy the working.

E3 $\$1.77$ from $\$10$

YOU DO Your turn.

13 $\$5.67$ from $\$50$ $\$44.33$

14 $\$6.09$ from $\$10$

15 $\$33.40$ from $\$50$

16 $\$10.52$ from $\$50$

17 $\$3.31$ from $\$10$

18 $\$8.73$ from $\$10$

19 $\$7.27$ from $\$20$

20 $\$3.20$ from $\$10$

21 $\$12.04$ from $\$20$

22 $\$14.63$ from $\$20$

23 $\$10.90$ from $\$20$

24 $\$3.82$ from $\$20$

25 $\$13.23$ from $\$20$

Medium

Questions 26–35

Find the total cost. Multiply, then write the answer in dollars and cents.

WE DO With your teacher: copy the working.

E1 6 items at $\$10.82$ each

E2 8 items at $\$7.68$ each

YOU DO Your turn.

26 6 items at $\$4.21$ each

$6 \times \$4.21$

$= \$25.26$

27 5 items at $\$5.42$ each

$5 \times \$5.42$

$= \underline{\hspace{2cm}}$

28 2 items at $\$12.25$ each

$\underline{\hspace{2cm}} \times \underline{\hspace{2cm}}$

$= \underline{\hspace{2cm}}$

29 9 items at $\$6.08$ each

30 9 items at $\$9.82$ each

31 7 items at $\$1.63$ each

32 4 items at $\$12.67$ each

33 5 items at $\$14.98$ each

34 8 items at $\$8.68$ each

35 6 items at $\$11.70$ each

Challenging

Questions 36–41

Solve the money problem. Show each step.

WE DO With your teacher: copy the working.

E1 You buy 3 drinks at $\$5.90$ and a sandwich at $\$5.01$. You pay with $\$50$. Find the change.

E2 Round $\$70.71$ to the nearest 5 cents for a cash payment.

YOU DO Your turn.

36 You buy 4 drinks at $\$2.21$ and a sandwich at $\$1.69$. You pay with $\$20$. Find the change.

$4 \times \$2.21 = \8.84

total: $\$8.84 + \$1.69 = \$10.53$

change: $\underline{\hspace{2cm}} - \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$

37 Round $\$60.66$ to the nearest 5 cents for a cash payment.

cents: $\underline{\hspace{2cm}} \rightarrow \underline{\hspace{2cm}}$

$= \underline{\hspace{2cm}}$

38 You buy 3 drinks at $\$4.56$ and a sandwich at $\$7.55$. You pay with $\$50$. Find the change.

39 Round $\$78.88$ to the nearest 5 cents for a cash payment.

40 You buy 3 drinks at $\$4.50$ and a sandwich at $\$7.60$. You pay with $\$50$. Find the change.

41 Round $\$16.90$ to the nearest 5 cents for a cash payment.